

Decay-rate variability by chemical compound

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Table ? Radioactive DecayRate Shifts Due to Chemical Compounding

Isotope (t1/2)

Form

??/?

Direction

Notes

?Be (53.22 d)

Be metal

?0.8%

Faster decay

Lower electron density around nucleus

?Be

BeF?

?0.9%

Faster decay

Ionic bond reduces local electron density

?Be

BeO

+0.5%

Slower decay

Higher electron density increases EC t1/2

?Be

Be(OH)?

+0.7%

Slower decay

Same mechanism as BeO

??Fe (2.737 y)

Fe(CN)???

?0.3%

Faster decay

Ligand field reduces selectron density

??Fe

Fe?O?

+0.2%

Slower decay

Oxide increases electron density

??Co (271.8 d)

CoCl?

?0.15%

Faster decay

Chloride complex reduces EC probability

??Co

Co metal

+0.1%

Slower decay

Metallic bonding increases electron density

??Co (70.86 d)

Co acetate

?0.2%

Faster decay

Ligand field effect

??Co

Co metal

+0.1%

Slower decay

Metallic bonding

??Zn (244 d)

ZnCl?

?0.2%

Faster decay

Ionic bonding reduces electron density

??Zn
 Zn metal
 +0.15%
 Slower decay
 Metallic bonding increases density
 ??Sr (50.5 d)
 SrCl?
 ?0.1%
 Faster decay
 Ionic bonding effect
 ??Sr
 SrCO?
 +0.1%
 Slower decay
 Carbonate increases electron density
 ???I (59.4 d)
 NaI
 ?0.2%
 Faster decay
 Ionic environment
 ???I
 I?
 +0.15%
 Slower decay
 Covalent bond increases electron density
 ???Ho (4570 y)
 HoCl?
 ?0.1%
 Faster decay
 Ligand field effect
 ???Ho
 Ho?O?
 +0.1%
 Slower decay
 Oxide increases electron density
 ???Yb (32 d)
 YbCl?
 ?0.1%
 Faster decay
 Ionic bonding
 ???Yb
 Yb?O?
 +0.1%
 Slower decay
 Oxide bonding
 ???Lu (1.37 y)
 LuCl?
 ?0.1%
 Faster decay
 Ionic bonding
 ???Lu
 Lu?O?
 +0.1%
 Slower decay
 Oxide bonding
 ???Au (186 d)
 AuCl?
 ?0.1%
 Faster decay
 Ligand field effect
 ???Au
 Au metal
 +0.1%
 Slower decay
 Metallic bonding
 ???Au (2.695 d)

AuCl?
 -0.1%
 Faster decay
 Same mechanism
 ???Au
 Au metal
 +0.1%
 Slower decay
 Same mechanism
 ???Hg (46.6 min)
 HgCl?
 -0.2%
 Faster decay
 Ionic bonding
 ???Hg
 Hg metal
 +0.1%
 Slower decay
 Metallic bonding
 ???Hg (46.6 d)
 HgCl?
 -0.2%
 Faster decay
 Ionic bonding
 ???Hg
 Hg metal
 +0.1%
 Slower decay
 Metallic bonding

Key Physical Interpretation

Chemical bonds change the orbital charge density at the nucleus by 10% to 10%.

Electron charge density shifts at the nucleus change the electron capture (EC) probability:

$\Delta EC \approx \Delta \rho_e(0) \cdot 2$.

The result is a decay rate variation between -0.9% to +0.7% (largest for lightest nucleus, ^7Be)

No other decay modes (α , β , γ) have ever shown reproducible chemical sensitivity.

References

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